

ENTROPY AND ENERGY: PARAMETER SENSITIVITY IN REACTION DIFFUSION DYNAMICS*

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Abstract. Reaction diffusion systems exhibit rich dynamical behaviors, including emergent pattern formation, phase transitions, and applications across science and engineering. This study evaluates the effectiveness of two well-documented measures for characterizing system dynamics: Shannon entropy, a statistical measure of information uncertainty, and the energy norm, a physical metric capturing both local and global deviations. A coupled reaction diffusion problem was solved across a range of diffusivity and reactivity parameters, producing solutions with diverse dynamical behaviors. Analysis of these solutions using both measures reveals their complementary strengths. The entropy measure excels in detecting sharp phase boundaries with minimal time data, while the energy norm provides detailed insights into subtle variations within dynamic regimes. These findings underscore the utility of combining these measures to achieve a comprehensive understanding of reaction diffusion dynamics.

1. Introduction. Phase transitions and criticality are central to understanding the emergent behavior of dynamical systems, appearing in a wide range of fields such as stochastic processes, data science, biological systems, and of course, statistical mechanics and thermodynamics. Historically, phase transitions have been studied through the lens of statistical mechanical thermodynamic models [1], where the system behavior is described by entropy S , energy E , and temperature T [2]. The interplay between these three quantities determines the systems evolution towards an equilibrium state x^* minimizing free energy G ,

$$x^* = \operatorname{argmin}_x \{G(x) = E(x) - TS(x)\}.$$

The dynamics of this system are widely applicable, and have been expanded beyond thermodynamics to information theory and complex systems analysis. Dynamics like those of energy and entropy often undergo phase transitions when the weighting parameter (temperature) of the two forces (energy and entropy) changes [3]. While temperature determines the phase, entropy is an excellent measure of phase changes, as sharp entropy gradients characterize critical points where emergent behavior changes [4]. Shannon entropy was originally introduced to quantify information uncertainty [5]. The generalization of entropy enabled applications in fields besides thermodynamics including data analysis and machine learning, file compression and transfer, quantum computing, and dynamical systems [6].

In dynamical systems, another approach to characterizing dynamic behavior has been to use the energy norm. This metric is effective as it captures both global deviations (via the L^2 norm) and local variations (via the gradient norm). Variations of the energy norm have been shown to measure behavior in thermoacoustic disturbances [7], fractures and mechanical degradation [7] [8], discontinuities in waveguides [9], and isotropic elastic wavefield imaging [10]. Together, these highlight the applicability of energy norms to characterize system dynamics.

Reaction diffusion problems [4] exhibit rich phase behavior and pattern formation, and are widely studied due to their emergent phenomena [11]. Alan Turing first used a model of coupled reaction diffusion equations to show how biological patterns emerge

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from an unstable process [12]. His research led to an understanding of how patterns such as stripes on a zebra can manifest from biochemical processes [11]. The emergent behavior of reaction diffusion systems is akin to those in thermodynamics, with sharp changes in system behavior as parameters vary.

This study will compare the effectiveness of a measure based on Shannon entropy and the energy norm in characterizing the dynamic behavior of a coupled reaction diffusion dynamical system. First, the specific reaction diffusion problem for analysis will be introduced. Second, the numerical scheme for solving the reaction diffusion problem will be outlined, and proofs of conditions required for accuracy will be presented. The energy and entropy measures will then be defined. Finally, the ability of the two measures to quantify dynamic behavior will be examined by using each to characterize the system for various points in the parameter space.

2. Problem. Consider the Schnakenberg model [13], a widely studied example of a reaction diffusion system on the square domain $\Omega = [0, 3] \times [0, 3]$ with two components u^1, u^2 :

$$\begin{cases} \frac{\partial u^1}{\partial t} = D_1 \Delta u^1 - R_1(u^1, u^2)u^1 + K_1 & \text{on } \Omega \times [0, t_{end}], \\ \frac{\partial u^2}{\partial t} = D_2 \Delta u^2 - R_2(u^1, u^2)u^2 + K_2 & \text{on } \Omega \times [0, t_{end}], \\ \frac{\partial u^i}{\partial x} = \frac{\partial u^i}{\partial y} = 0 & \text{on } \partial\Omega \text{ for } i = 1, 2. \end{cases}$$

Here, D_1 and D_2 represent diffusion coefficients and R_1, R_2 define the reactivity of each component. Adapting the dimensionless version of the system [11], define the R functions and K constants as follows:

$$\begin{aligned} R_1(u^1, u^2) &= \gamma - \gamma u^1 u^1, \\ R_2(u^1, u^2) &= \gamma (u^1)^2, \\ K_1 &= \gamma a, \\ K_2 &= \gamma b. \end{aligned}$$

The constants $a = 0.1305$ and $b = 0.7695$ were fixed according to instability conditions [14]. The initial conditions of the system were small perturbations from steady state of $u^{(1)}$ [15]. Steady state values were derived by setting the time derivatives to zero, and when the small Gaussian perturbation is added to $u^{(1)}$, the initial conditions are:

$$\begin{aligned} u^1(x, y, 0) &= a + b + 10^{-3} e^{-100((x-\frac{3}{2})^2 + (y-\frac{3}{2})^2)}, \\ u^2(x, y, 0) &= \frac{b}{(a+b)^2}. \end{aligned}$$

By then defining the diffusion ratio $d = D_2/D_1$, the dynamic behavior of the solution is characterized solely by the diffusion ratio d and the reactivity constant γ . In other words, the behavior of solutions are defined by a single point in the parameter space $(\gamma, d) \in \mathcal{P}$. In the thermodynamics analogy, this point is akin to the state variables of temperature and pressure, which characterize the equilibrium phase (emergent behavior) of the solution. As such, this study will solve the problem for a variety of points in $\mathcal{P}$ and test how Shannon entropy and the energy norm can indicate emergent properties, just like classical entropy and energy does in thermodynamics.

3. Numerical solution scheme. To solve the problem, the finite element method is used for the spatial piece, and a modified Implicit Euler two level finite difference scheme is used for the temporal component. Examine the generic reaction diffusion problem

$$\frac{\partial u}{\partial t} = D\Delta u - R(u^i)u + K,$$

where u_i are any coupled terms contributing to the reactivity term. The weak form of the reaction diffusion equation is derived by multiplying by a test function $v \in H_0^1(\Omega)$ and integrating over the domain Ω

$$\int_{\Omega} \frac{\partial u}{\partial t} v \, d\Omega = \int_{\Omega} D\Delta uv \, d\Omega - \int_{\Omega} R(u^i)uv \, d\Omega + \int_{\Omega} Kv \, d\Omega.$$

After integrating by parts, the boundary integrals cancel due to the no flux Neumann boundary conditions. This leaves the weak form as

$$\int_{\Omega} \frac{\partial u}{\partial t} v \, d\Omega + \int_{\Omega} D\nabla u \cdot \nabla v \, d\Omega + \int_{\Omega} R(u^i)uv \, d\Omega = \int_{\Omega} Kv \, d\Omega.$$

To implement the Implicit Euler finite difference scheme for the temporal portion of the equation, approximate the temporal derivative as

$$\frac{\partial u}{\partial t} \approx \frac{u_n - u_{n-1}}{h_t},$$

where h_t is the time step size, and u_{n-1} is the previous time's solution. To maintain linearity in the spatial portion, the reactivity is based on the previous time step. Under the implicit scheme, the u functions are replaced by u_n . After multiplying through by h_t we have the final weak form of problem:

$$\int_{\Omega} u_n v \, d\Omega + h_t \left(\int_{\Omega} D\nabla u_n \cdot \nabla v \, d\Omega + \int_{\Omega} R(u_{n-1}^i)u_n v \, d\Omega \right) = \int_{\Omega} u_{n-1} v \, d\Omega + \int_{\Omega} Kv \, d\Omega.$$

Linear triangular basis elements were used to approximate the solution. Simpson's quadrature routine [16] was used to approximate the integrals of the weak form. Three quadrature points (x_q, y_q) were used for the reaction term due to its nonlinearity, with a single point used for all other integrals. Since they are time independent, mass M and diffusion D matrices were assembled and stored using the basis elements using

$$M_{i,j} = \sum_q w_q \phi_i(x_q) \phi_j(y_q),$$

$$D_{i,j} = h_t D \sum_q w_q \nabla \phi_i(x_q) \cdot \nabla \phi_j(y_q),$$

To iterate one time step, the time dependent reaction matrix $R^{(n)}$ and load vector $\mathbf{f}^{(n)}$ were assembled using the previous solutions for each coupled element of the system $u_{n-1}^{(i)}$ as follows:

$$R_{i,j}^{(n)} = \sum_q w_q R(u_{n-1}^i(x_q, y_q)) \phi_i(x_q) \phi_j(y_q),$$

$$\mathbf{f}_i^{(n)} = \sum_q w_q (u_{n-1}(x_q) + K) \phi_i(x_q).$$

The resulting linear system,

$$(M + D + R^{(n)})u_n = \mathbf{f}^{(n)},$$

was solved using a python algebraic multigrid package [17]. A spatial step size $h_x = h_y = 0.06$ was used, creating a 50×50 node spatial grid on which solutions were approximated. A time step size of $h_t = 0.01$ was used to evaluate 30 time points (ending with $t = 0.3$).

3.1. Stability and convergence of solution scheme. To show convergence and stability of solutions approximated in $\mathcal{V} = H_0^1(\Omega)$, the bilinear and linear form of the weak formulation of the reaction diffusion problem will be shown to have key properties. The bilinear form a and linear form l are given by

$$a(u_n, v) = \int_{\Omega} u_n v \, d\Omega + h_t \left(\int_{\Omega} D \nabla u_n \cdot \nabla v \, d\Omega - \int_{\Omega} R(u_{n-1}) u_n v \, d\Omega \right), \quad (3.1)$$

$$l(v) = \int_{\Omega} u_{n-1} v \, d\Omega + \int_{\Omega} K v \, d\Omega. \quad (3.2)$$

PROPOSITION 3.1. *The linear functional $l(v)$ is bounded.*

Proof. To show that the linear form is bounded, find a constant $C > 0$ such that

$$|l(v)| \leq C \|v\|_{H_0^1}.$$

By Hölder's inequality and the definition of the Sobolev norm,

$$\left| \int_{\Omega} (u_{n-1} + K) v \, d\Omega \right| \leq \|u_{n-1} + K\|_2 \|v\|_2 \quad (3.3)$$

$$\leq \|u_{n-1} + K\|_2 (\|v\|_2 + \|\nabla v\|_2) = C \|v\|_{H_0^1}. \quad (3.4)$$

□

PROPOSITION 3.2. *The bilinear form $a(u, v)$ is continuous provided that $R(u_{i,n-1}) \leq R_{max}$ and $1 + h_t R_{max} > 0$.*

Proof. To show $a(u, v)$ is continuous, find a constant $C > 0$ such that

$$a(u, v) \leq C \|u\|_{H_0^1} \|v\|_{H_0^1}.$$

Factoring out the lower bound R_{max} and reorganizing we have

$$a(u, v) \leq (1 + h_t R_{max}) \int_{\Omega} uv \, d\Omega + h_t D \int_{\Omega} \nabla u \cdot \nabla v \, d\Omega.$$

Next, apply the Cauchy-Schwartz inequality to get

$$a(u, v) \leq (1 + h_t R_{max}) \|u_n\|_2 \|v\|_2 + h_t D \|\nabla u_n\|_2 \|\nabla v\|_2.$$

Manipulating the right hand side of the continuity condition,

$$C \|u\|_{H_0^1} \|v\|_{H_0^1} = C (\|u\|_2 \|v\|_2 + \|u\|_2 \|\nabla v\|_2 + \|\nabla u\|_2 \|v\|_2 + \|\nabla u\|_2 \|\nabla v\|_2), \quad (3.5)$$

$$= C (\|u\|_2 \|v\|_2 + \|\nabla u\|_2 \|\nabla v\|_2) + C_2, \quad (3.6)$$

where $C_2 = C (\|u\|_2 \|\nabla v\|_2 + \|\nabla u\|_2 \|v\|_2) \geq 0$. By choosing C such that

$$C \geq \max(1 + h_t R_{max}, h_t D) > 0,$$

Continuity of the bilinear form is shown with

$$a(u, v) \leq \max(1 + h_t R_{max}, h_t D) (\|u\|_2 \|v\|_2 + \|\nabla u\|_2 \|\nabla v\|_2) \leq C \|u\|_{H_0^1} \|v\|_{H_0^1}$$

□

PROPOSITION 3.3. *The bilinear form $a(u, v)$ is coercive provided that $R_{min} \leq R(u_{n-1})$ and $1 + h_t R_{min} > 0$.*

Proof. To show that $a(u, v)$ is coercive, find a constant $C > 0$ such that

$$a(u, u) \geq C \|u\|_{H_0^1}^2.$$

Bounding the bilinear form from below using R_{min} results in

$$a(u, u) \geq (1 + h_t R_{min}) \int_{\Omega} u^2 d\Omega + h_t D \int_{\Omega} \nabla u^2 d\Omega = (1 + h_t R_{min}) \|u\|_2^2 + h_t D \|\nabla u\|_2^2.$$

By choosing C such that

$$0 < C \leq \min(1 + h_t R_{min}, h_t D),$$

coercivity follows with the bound

$$a(u, u) \geq \min(1 + h_t R_{min}, h_t D) (\|u\|_2^2 + \|\nabla u\|_2^2) \geq C \|u\|_{H_0^1}^2.$$

□

THEOREM 3.4. *Solutions of the reaction diffusion problem are unique and converge to the smooth solution u provided the conditions for coercivity and ellipticity are met.*

Proof. By the Lax-Milgram Theorem [18], since $a(u, v)$ is continuous and coercive and $l(v)$ is a bounded linear functional, there exists a unique $u \in \mathcal{V}$ which solves the weak form. Define $\mathcal{T}$, a piecewise linear triangular mesh on which the solution will be approximated. Since $\mathcal{T}$ is within the family of non-degenerate simplex meshes on a polyhedral domain $\Omega_h \subset \mathbf{R}^n$, the error of the approximate solution u_h on this mesh is bounded [18]. Temporal stability follows from spatial stability, since each time step is an instance of solving the stable spatial system which depends on the previous time step. □

4. Characterization of parameter space. This section will examine the system dynamics across the parameter space, and characterize the dynamics using the entropy and energy measures. The advantages and disadvantages of each measure will be assessed in their ability to capture the impact of changing γ and d . Then, phase plots of the parameter space will be presented and discussed. Finally, the solution behavior in different regimes will be visualized to show characteristics of different regimes.

The measures being examined are Shannon entropy and the energy norm, defined as follows:

$$s(x) = - \sum_{i,j} \hat{x}_{i,j} \log \hat{x}_{i,j}, \quad (4.1)$$

$$e(x) = \|x\|_E = \|\nabla x\|_2 + \|x\|_2. \quad (4.2)$$

Here, $\hat{x}$ denotes a vector that sums to one, or the probability density form of x .

4.1. Impact of γ and d . For each $(\gamma, d) \in \mathcal{P}$ we computed the entropy $s(u, t)$ and energy $e(u, t)$ of u^1 along the trajectory. At each time we then formed the (marginal) squared sensitivities

$$\left(\frac{\partial m(t)}{\partial \gamma}\right)^2, \quad \left(\frac{\partial m(t)}{\partial d}\right)^2, \quad m \in \{s, e\},$$

and averaged them over the parameter grid. The resulting curves are shown in Fig. 4.1.

Entropy is consistently more sensitive to *both* parameters at early times and, on average, carries *more information about* γ than energy across the time window. Its dependence on d is also stronger than its dependence on γ , reflecting that entropy primarily tracks dispersion. Energy’s sensitivities grow with time; it overtakes entropy for γ only at later time points, as amplitudes increase and gradients sharpen. The gaps in the energy panel indicate time indices we masked because extremely large magnitudes made numerical derivatives unreliable.

For inference at early times, entropy is the most informative observable for both d and γ ; energy becomes competitive (and for γ , superior) only later, with the caveat of occasional unusable time points.

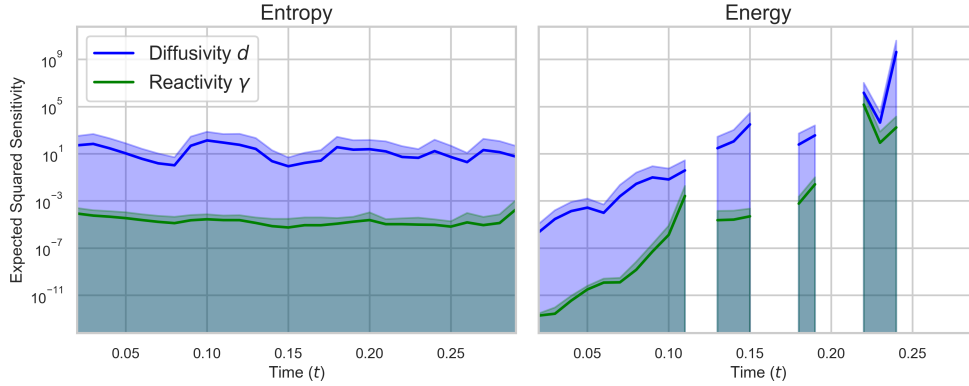


FIG. 4.1. *Expected squared sensitivities for entropy (left) and energy (right), averaged over $(\gamma, d) \in \mathcal{P}$: blue— $\mathbf{E}[(\partial m/\partial d)^2]$, green— $\mathbf{E}[(\partial m/\partial \gamma)^2]$. Shading shows the confidence interval.*

4.2. Phase plots.

4.3. Phase plots. To summarize the parameter-dependent behavior under each measure, we integrate the entropy $s(u^1, t)$ and energy $e(u^1, t)$ over an early-time window. We restrict to the first 10 time steps (up to $t = 0.10$), where the energy gradients remain numerically stable. The values for the heatmap phase plots in Fig. 4.2 shown at each (γ, d) are

$$M(\gamma, d) = \int_0^{0.10} m(u^1(t)) dt \approx \sum_{n=1}^{10} m(u_n^1) h_t, \quad m \in \{s, e\},$$

with u_n^1 the solution at time t_n and h_t the time step.

Entropy defines a more informative geometry over the parameter space by exhibiting strong sensitivity to changes in reactivity γ , particularly at early times. This results in a clear partitioning of the parameter domain into distinct regimes, enabling

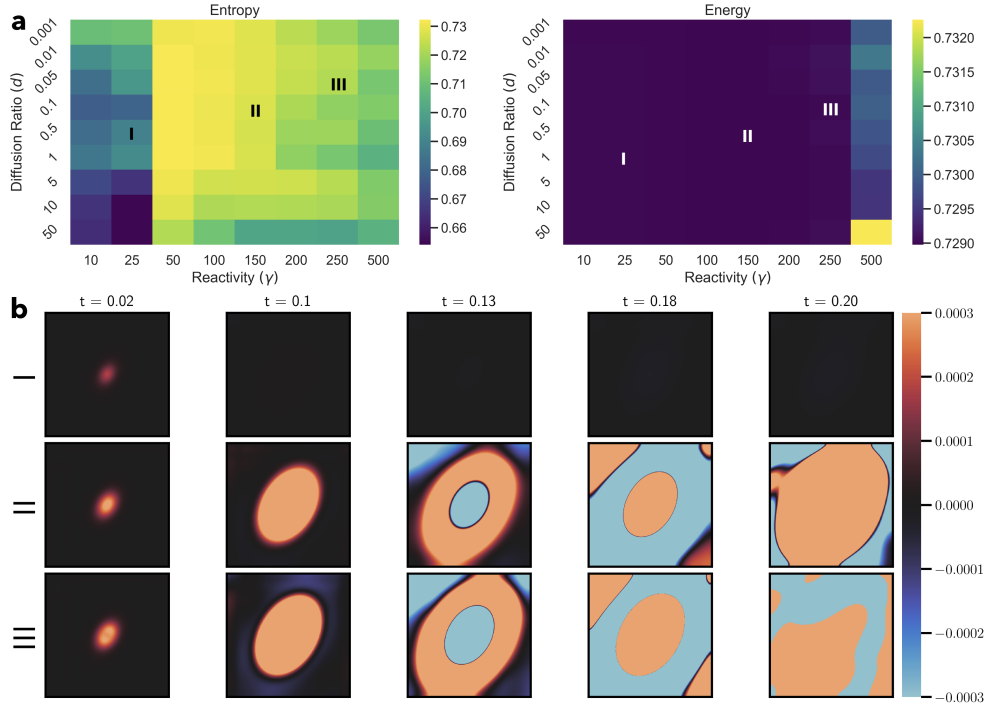


FIG. 4.2. (a) Heatmap phase plots of the integrated energy and entropy measures for points $(\gamma, d) \in \mathcal{P}$. (b) Numerical solutions of u^1 for three representative points in the parameter space: I ($\gamma = 10, d = 0.5$), II ($\gamma = 150, d = 0.5$), and III ($\gamma = 150, d = 0.1$).

early detection of phase transitions as seen in the sharp boundary of the entropy map between $\gamma = 25$ and 50.

In contrast, energy remains largely insensitive across most of the parameter space during early times, showing minimal variation except in regions with very high γ . The informational advantage exhibited in Fig. 4.1 is clearly reflected here. The entropy phase plot not only separates the parameter space into regimes but also shows subtle gradients within each regime, suggesting that entropy is sensitive and informative for both γ and d . It is possible that with more reactivity values sampled between $\gamma = 250$ and 500, the energy norm would demonstrate greater sensitivity earlier in time. Additionally, the limited resolution of the energy plot may obscure finer transitions that could occur within the early-time window. With denser sampling, energy might better capture the onset of divergence. Nonetheless, at the current resolution, entropy clearly provides richer information across a broader region of parameter space, detecting both phase boundaries and within-phase variation. Energy, by contrast, serves primarily as a late-time indicator of instability, sharply increasing only in the most reactive regimes. Together, these results support the conclusion that entropy is the more effective measure for early identification of dynamic transitions, while energy captures extreme behaviors emerging later in time.

Three points in parameter space $\{I = (10, 0.5), II = (150, 0.1), III = (250, 0.05)\} \in \mathcal{P}$ are marked in Fig. 4.2a. The solutions for each of these points are visualized in Fig. 4.2b, illustrating the system's response under varying reactivity γ and diffusion ratio d . At the first point (I), where $\gamma = 10$ and $d = 0.5$, the perturbation from steady

state is quickly smoothed by the diffusion dominated dynamics. The strong diffusion prevents the growth of spatial gradients, and the system quickly returns to steady state by $t = 0.1$. The solution represents a stable and smooth recovery to equilibrium steady state.

At the second point (II), where $\gamma = 150$ and $d = 0.1$, the system transitions to a chaotic state. The initial perturbation triggers the periodic generation of well-defined, ripple-like structures that expand symmetrically. This behavior arises from the pattern-forming nature of reaction-diffusion systems, where reactivity promotes spatial gradients while diffusion smooths them. Over time, the ripples amplify in both magnitude and spatial extent, increasing disorder until the system begins to diverge significantly from the steady state. The structured rings highlight a balance between diffusion and reactivity.

The third point (III), where $\gamma = 250$ and $d = 0.05$, has similar, yet accelerated dynamics compared to point II. The higher reactivity overwhelms the limited diffusive effects, leading to a rapid amplification of small perturbations. The periodic ripples grow in magnitude and propagate faster, eventually breaking down into large, irregular patterns. The dominance of reactivity in this regime drives the system into a chaotic and disordered state far more quickly, with diffusion unable to regulate the reactive forces.

5. Discussion. This study compared the effectiveness of two scalar observables—Shannon entropy and the energy norm—in characterizing the behavior of a coupled reaction–diffusion system across a range of reactivity (γ) and diffusion ratio (d) values. Both measures captured system divergence from equilibrium, but with distinct sensitivities and informational advantages depending on the parameter regime and timescale.

The entropy measure was highly effective at detecting early transitions in system behavior. It exhibited strong sensitivity to changes in reactivity γ , allowing rapid identification of phase boundaries with minimal time data. In particular, entropy delineated a sharp transition between stable and unstable regimes around $\gamma = 25$ –50, with additional sensitivity to variations in diffusivity d within each regime. Over time, entropy provided consistent and smooth growth patterns, making it well suited for early-stage analysis and parameter estimation.

In contrast, the energy norm remained nearly constant across most of the parameter space during early times, only becoming sensitive at the highest values of γ . Its limited variation at early time points restricted its usefulness for early detection of transitions. However, the energy norm became informative at later times, capturing blow-up behavior and rapid amplification in unstable regimes. This complementary sensitivity highlights energy as a late-time diagnostic of instability, whereas entropy excels at early-time phase detection.

One limitation encountered was the numerical instability in computing the energy norm at later time steps. Sharp gradients and diverging magnitudes caused unreliable derivative estimates in some regions of parameter space. Developing a more robust numerical solution scheme—for example, by employing higher-order spatial discretization or stabilized time stepping—would allow the energy norm to be evaluated more smoothly and extend its utility for long-term analysis.

6. Conclusion. Overall, entropy and energy offer complementary insights into the dynamics of reaction–diffusion systems. Entropy is a sensitive early indicator of reactivity-driven transitions, while energy captures extreme behaviors associated with instability and divergence. These findings have practical implications for parameter

estimation and phase identification in chemical and biological systems, where rapid detection of instability or pattern formation is critical. Together, the two measures form a versatile framework for analyzing complex spatiotemporal behavior, with potential applications in reaction engineering, materials science, and biological pattern formation. Future work may extend these results by integrating both measures into a combined free energy framework, which could enhance sensitivity across time scales and parameter regimes.

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REFERENCES

- [1] Nigel Goldenfeld. *Lectures on Phase Transitions and the Renormalization Group*. Addison-Wesley, 1992.
- [2] Herbert B. Callen. *Thermodynamics and an Introduction to Thermostatistics*. Wiley, 1985.
- [3] Mehran Kardar. *Statistical Physics of Fields*. Cambridge University Press, 2007.
- [4] M. C. Cross and P. C. Hohenberg. Pattern formation outside of equilibrium. *Reviews of Modern Physics*, 65(3):851, 1993.
- [5] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.
- [6] David J. C. MacKay. *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press, 2003.
- [7] K. J. George and R. I. Sujith. Disturbance energy norms: A critical analysis. *Journal of Sound and Vibration*, 2012.
- [8] M. V. Pham, M. N. Nguyen, and T. Q. Bui. Numerical simulation of localized quasi-brittle fracture with an enhanced bi-energy norm based equivalent strain. *Engineering Fracture Mechanics*, 2023.
- [9] J. C. Simo and J. W. Ju. Strain-and stress-based continuum damage models—i. formulation. *International Journal of Solids and Structures*, 1987.
- [10] L. E. Garcia-Castillo, D. Pardo, and I. Gomez-Revuelto. A two-dimensional self-adaptive hp finite element method for the characterization of waveguide discontinuities. part i: Energy-norm based automatic hp-adaptivity. *Computer Methods in Applied Mechanics and Engineering*, 2007.
- [11] J. D. Murray. *Mathematical Biology II: Spatial Models and Biomedical Applications*. Springer, 2003.
- [12] Alan Turing. The chemical basis for morphogenesis. *Philosophical Transactions of the Royal Society B*, 1952.
- [13] Fatemeh Shakeri and Mehdi Dehghan. The finite volume spectral element method to solve turing models in the biological pattern formation. *Computers and Mathematics with Applications*, 2011.
- [14] S.A. Newman M. Alber J. Zhu, Y.-T. Zhang. Application of discontinuous galerkin methods for reaction–diffusion systems in developmental biology. *Journal of Scientific Computing*, 2009.
- [15] J.G. Verwer W. Hundsdorfer. Numerical solution of time-dependent advection–diffusion–reaction equations. *Springer, Berlin*, 2003.
- [16] Kendall E. Atkinson. *An Introduction to Numerical Analysis*. Wiley, 2nd edition, 1989.
- [17] Nathan Bell, Luke N. Olson, and Jacob Schroder. Pyamg: Algebraic multigrid solvers in python. *Journal of Open Source Software*, 2024.
- [18] Scott MacLachlan James Adler, Hans De Sterck and Luke Olson. *Numerical Partial Differential Equations*. N/A, preprint edition, 2024.
- [19] Andrei A. Klishin, Joseph Bakarji, J. Nathan Kutz, and Krithika Manohar. Statistical mechanics of dynamical system identification, 2024.